

Week 3 作业——完整提交版 PDF

生物信息学——第 3 周作业

公共数据库的数据获取、清洗与质量控制 (GSE146853)

本 PDF 由以下内容整合而成：

- 1) 主文档 Week3_Homework.pdf (本 PDF 第 2 - 9 页, 共 8 页) : 两题完整作答、Q
- 2) 附录 1: week3_homework.R (修复版 R 脚本原文)
- 3) 附录 2: cleaning_summary.tsv (清洗日志)
- 4) 附录 3: ai_verification_log.txt (AI 使用记录)
- 5) 附录 4: library_size_summary.tsv (文库大小明细)
- 6) 附录 5: sessionInfo.txt (R 环境记录)

生物信息学——第3周作业

一、第一题：Week 2 数据驱动问题与数据集分诊

1.1 Week 2 数据驱动问题

在 IBS 患者及健康对照的肠黏膜活检样本中，IBS-D（腹泻型）与健康对照之间的宿主转录组表达谱是否存在差异？哪些基因和通路与 IBS-D 表型相关？数据类型为 bulk RNA-seq gene-count matrix。

1.2 确认研究身份

项目	信息
GEO 系列 (Series)	GSE146853
标题	Longitudinal multi-omics reveals subset-specific mechanisms underlying irritable bowel syndrome
物种	Homo sapiens
主要发表文献	Mars et al., Cell, 2020, PMID 32916129
Week 2 相关论文	Priya et al., Nature Microbiology, 2022; 同一 IBS 队列的复用分析
平台	GPL16791; Illumina HiSeq 2500
测序/分析信息	MAPR-Seq; 参考基因组 hg19
GEO 样本数	68

1.3 Accession 映射

层级	编号	说明
研究	GSE146853	GEO 研究级编号
样本	GSM4407881	示例样本
受试者	10007589-1	受试者/时间点信息
生物样本	SAMN14364283	NCBI BioSample
SRA 实验	SRX7899630	示例样本对应 SRA experiment
生物项目	PRJNA612180	研究关联 BioProject
SRA 项目	SRP252532	原始测序数据项目

GSM4407881 为肠黏膜活检 RNA-seq 样本，cohort 为 Constipation；对应 BioSample SAMN14364283 和 SRA experiment SRX7899630。

1.4 可用数据层

数据层	文件/入口	采用情况
系列矩阵	GSE146853_series_matrix.txt.gz	用于元数据；不作为 raw counts 表达矩阵
补充计数文件	GSE146853_GeneCount_raw.txt.gz (约5.0 MB)	本作业主要分析层
原始测序读段	SRA SRP252532 / FASTQ	可用，但68个样本重新比对超出一小

		时作业范围
单细胞文件	未发现	本研究为 bulk RNA-seq

1.5 研究设计重建

设计要素	重建结果
总体设计	纵向研究；同一受试者存在重复观察，不能默认样本相互独立
受试者	42 位受试者；GEO 系列共 68 个转录组样本
主要 cohort	IBS-C 23；IBS-D 21；Healthy 21
元数据修正	2 例 “Diarrheaiarrhea” 已核对并统一为 Diarrhea；1 个 cohort 缺失样本从主对比排除
主对比	T1: Diarrhea 12 vs Healthy 13, n=25
重复测量	保留 patient_id + timepoint
主要协变量	age、sex、BMI、cohort
批次	公开元数据无明确 batch 字段，无法充分评估，记录为局限

1.6 Dataset Triage Score

标准	得分	依据
生物学匹配	2/2	IBS 肠黏膜宿主转录组与研究问题直接匹配
重复性	2/2	68 个转录组样本，包含多个生物学受试者/组
元数据	2/2	有 cohort、patient/time-point、人口学字段；batch 仍需注意
文件	2/2	raw gene-count matrix + SRA 链接
可追溯性	2/2	论文、GEO、平台和 accession 链条可追溯
获取与伦理	2/2	公开数据，可用于本课堂流程
总分	12/12	继续但有局限（Proceed with limitations）

决定：继续但有局限（Proceed with limitations）。数据集与问题高度匹配且数据可追溯，但纵向重复测量结构及 batch 信息缺失限制了直接比较方式；后续分析必须预先定义 contrast，并保留 subject/time-point 信息。

数据获取与可重复性记录

字段	记录
编号 (Accession)	GSE146853
主文件 (Primary file)	GSE146853_GeneCount_raw.txt.gz
文件类型 (File type)	gzip 压缩、制表符分隔的原始基因计数矩阵
文件大小 (Approx. size)	约 5.0 MB
获取方式 (Retrieval)	R / GEOquery: getGEO() + getGEOSupFiles()
项目结构	data_raw / metadata / R / data_clean / results
原始数据原则	原始文件不修改；所有清洗和结果写入新路径

本作业使用 count matrix 而非 FASTQ，是因为 Week 3 的重点是数据获取、检查、清洗、metadata 对齐和 QC，而不是完整 raw-read 重处理。

完整性检查与清洗摘要

检查	结果/规则	R证据
矩阵维度	64,253 genes × 68 samples	dim(counts)
计数类型 (Counts)	非负整数	all(x >= 0); all(x == round(x))
缺失值	显式检查，不自动插补	sum(is.na(x))
重复特征 (feature)	检查 GeneID	anyDuplicated(rownames(counts))
重复样本 (sample)	检查样本列名	anyDuplicated(colnames(counts))
样本对齐	counts 与 metadata 显式映射后再分析	identical(colnames(counts), metadata\$sample_id)
过滤规则	至少在最小分析组样本中 count ≥10	rowSums(x >= 10) >= smallest_group
过滤前后	64,253 → 约24,204 genes	按上述规则过滤

清洗原则：不把 TPM/FPKM 当作 raw counts；不通过 round() 将连续表达值伪装成 counts；低表达过滤仅是本次分析准备规则，不是普适的唯一阈值。

QC 图

本版 QC 图由附录 R 脚本实际运行生成：脚本从 GEO 下载并读取真实 count matrix，完成样本对齐与低表达过滤后输出下图 (results/GSE146853_library_size.png)，图中的 25 个 T1 样本与第 1.5 节定义的分析子集一致。

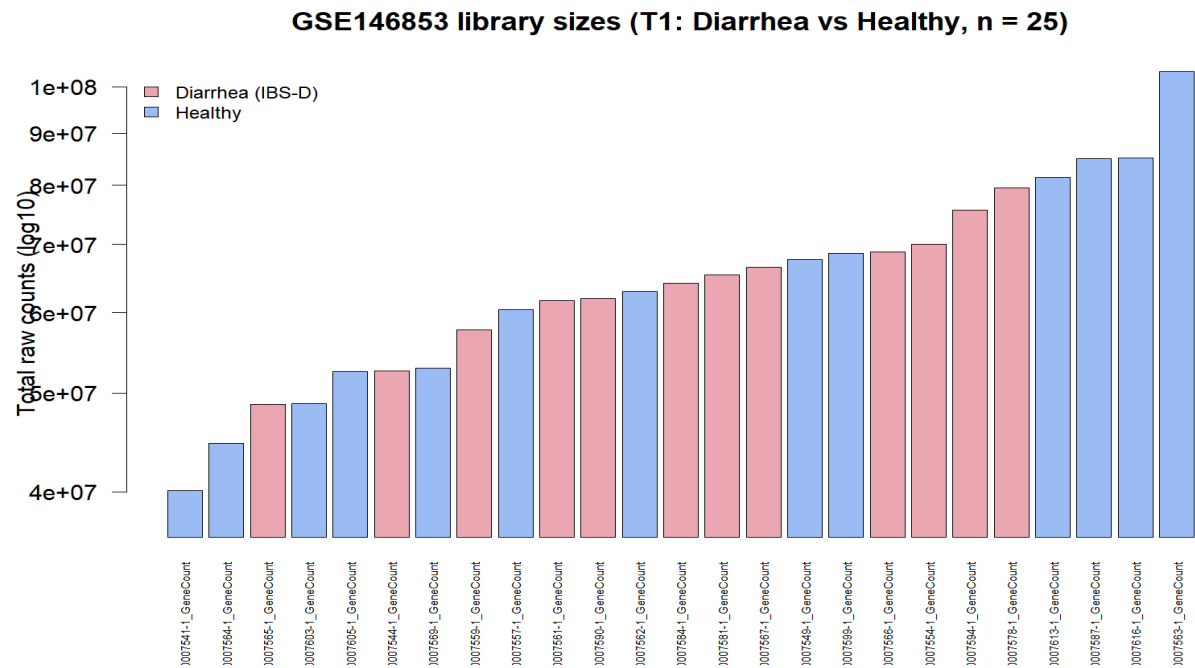


图1. GSE146853 T1 时点 Diarrhea (IBS-D) 与 Healthy 样本的 library size (n = 25) 。

QC 解读：25 个样本的总 raw counts 约分布在 $4 \times 10^7 - 1 \times 10^8$ 范围内，样本间测序深度存在一定差异，但未见明显极端低深度样本。Diarrhea 与 Healthy 样本的 library sizes 整体交错分布，没有看到由分组造成的明显测序深度偏倚；不过 library size 只能反映测序深度，不能替代 PCA 或样本相关性分析。

AI 辅助与验证记录

项目	记录
提示词	使用第 3 周课件，帮我重建 GSE146853 的 GEO accession 链条与分析就绪设计；识别 GSE、GSM、BioProject/SRA 链接、平台、数据层、样本结构与合适的清洗/QC 检查；不要编造缺失的元数据。
采纳的建议	区分 GSE/GSM/GPL/BioProject/SRA；先检查 supplementary counts；验证非负整数；保留 patient ID/time point；先定义 contrast；raw files 不修改。
核验	GEO GSE146853；GSM4407881；R checks：非负、整数、NA、重复 ID、metadata/sample 对齐。
拒绝的建议	不把所有样本默认视为独立重复；不对 TPM/FPKM 直接运行 DESeq2；R 未实际运行前不声称 QC 已完成。

反思

GSE146853 最主要的局限在于其纵向研究设计：同一受试者的多次活检不能被视为相互独立的观测。该数据集可以支持“在预先定义的 IBS 亚组与健康对照之间，宿主转录组模式存在或不存在差异”这一判断，但仅凭它本身无法在宿主基因表达与肠道微生物组之间建立因果关系。后续分析应保留受试者层面的配对关系与时间点信息，再进行基于计数的差异表达分析与通路富集分析。一个有用的验证步骤是：在独立的 IBS 转录组队列中检验最强的候选基因或通路，或通过针对性的实验加以验证。

二、第二题：GSE87487 Mini-case

GSE87487 是人类 bulk RNA-seq 研究，包含 20 个 liver allograft biopsies，在 pre-reperfusion 与 post-reperfusion 取样，并提供约 3.7 MB supplementary count matrix；关联 SRA project SRP090633。

问题	学生判断	依据
2.1 这些值是 counts 吗？	是；需核验整数性	文件为 GSE87487_counts.20samples.txt.gz，方法使用 featureCounts；下载后检查非负整数
2.2 样本独立吗？	不独立	10 位供体各有 pre/post 配对活检，必须保留 subject pairing
2.3 能直接比较全部20个样本吗？	不能	IRI status 与 pre/post-reperfusion 构成多个因素，必须先定义具体 contrast
2.4 能否完整重处理 raw reads？	可以，但本作业不做	SRA SRP090633 提供 raw data，但超出一小时范围

结论

GSE146853 的数据集分诊为 12/12，决定为“继续但有局限”。它提供可追溯的 GEO/SRA accession、bulk RNA-seq raw count matrix 和可用的样本元数据；主要限制是纵向重复测量和公开元数据缺少明确 batch 字段。Week 3 的分析边界应停留在获取、完整性检查、样本对齐、低信息基因过滤和 QC，不提前完成正式差异表达分析。本次实际生成的 library-size QC 图显示 25 个 T1 样本的测序深度总体可接受，未见明显极端低深度样本。

参考资料与核验来源

- Priya S. et al. (2022). Nature Microbiology. Identification of shared and disease-specific host gene-microbiome associations across human diseases using multi-omic integration.
- Mars R.A.T. et al. (2020). Cell. Longitudinal multi-omics reveals subset-specific mechanisms underlying irritable bowel syndrome. PMID: 32916129.
- NCBI GEO: GSE146853 — Longitudinal multi-omics reveals subset-specific mechanisms underlying irritable bowel syndrome.
- NCBI GEO sample: GSM4407881.
- NCBI BioProject / SRA: PRJNA612180; SRP252532.
- 课程提供的第 3 周作业与阅读材料 (Week 3 Homework & Week 3 Reading Material)。

附录 A: R 脚本 (week3_homework.R)

```
# week3_homework.R — GSE146853 (修正版, 可直接运行)
# 1) read.delim 用 row.names = "GeneID" (不要用第 1 列 Chr),
# 并只保留 68 个 s_* 计数列 (初版会把 6 列注释也当数据)。
# 2) 计数列名是 s_{患者}-{时间点}_GeneCount, 不是 GSM 编号;
# 通过 meta$title 建立 列名 -> GSM 的显式映射 (初版 intersect 会得 0)。
# 3) patient_id / timepoint 不在 characteristics 里, 而在 title
# (如 10007589-1); characteristics 只有 age/bmi/gender/cohort。
# 4) 先把分析子集定义为 T1 的 Diarrhea vs Healthy, 再算最小分组。
# =====
# 若 GEOquery 未安装才需执行 (本机已装; 勿运行 install, 否则会触发 Update all/some/none 询问):
# if (!requireNamespace("BiocManager", quietly=TRUE)) install.packages("BiocManager")
# BiocManager::install("GEOquery", update = FALSE, ask = FALSE)
library(GEOquery)

dir.create("data_raw/GSE146853", recursive = TRUE, showWarnings = FALSE)
dir.create("metadata", recursive = TRUE, showWarnings = FALSE)
dir.create("results", recursive = TRUE, showWarnings = FALSE)
dir.create("data_clean", recursive = TRUE, showWarnings = FALSE)

# ---- 1. GEO 研究对象 (Series Matrix 主要取元数据) ----
gse_list <- getGEO("GSE146853", GSEMatrix = TRUE, getGPL = FALSE)
se <- gse_list[[1]]
if (inherits(se, "ExpressionSet")) {
  meta <- Biobase::pData(se)
} else {
  meta <- as.data.frame(colData(se))
}
```

```

cat("metadata rows:", nrow(meta), "\n")
print(colnames(meta)) # 先看有哪些列 (title / geo_accession / characteristics...)

# ---- 2. 补充文件清单 -> 下载计数矩阵 ----
# 注: getGEOSuppFiles() 的参数是 fetch 而非 fetch_files, 仅下载即可
getGEOSuppFiles("GSE146853", baseDir = "data_raw")
count_file <- list.files("data_raw/GSE146853",
                        pattern = "GSE146853_GeneCount_raw\\.txt\\.gz$",
                        full.names = TRUE)

stopifnot(length(count_file) == 1)

# ---- 3. 读入: GeneID 作行名; 只留 68 个计数列 ----
counts_all <- read.delim(count_file, row.names = "GeneID",
                        check.names = FALSE, stringsAsFactors = FALSE)
counts <- counts_all[, grepl("^s_", colnames(counts_all))]
dim(counts) # 预期 64253 x 68

# ---- 4. 计数性质核验 (全部应通过) ----
stopifnot(all(as.matrix(counts) >= 0)) # 非负
stopifnot(all(as.matrix(counts) == round(as.matrix(counts)))) # 整数
stopifnot(sum(is.na(counts)) == 0) # 无缺失
stopifnot(anyDuplicated(rownames(counts)) == 0) # GeneID 无重复
range(counts) # 0 ~ 4184346

# ---- 5. 从标题解析患者/时间点, 从 characteristics 解析 cohort ----
cnt_title <- sub("_GeneCount$", "", sub("^s_", "", colnames(counts))) # "10007541-1"
m <- match(cnt_title, meta$title)
stopifnot(!any(is.na(m))) # 68/68 必须全部匹配

design <- data.frame(
  sample_id = colnames(counts),
  title = cnt_title,
  gsm = meta$geo_accession[m],
  patient_id = sub("-[12]$", "", cnt_title),
  timepoint = sub("[0-9]+-", "", cnt_title), # "1" 或 "2"
  stringsAsFactors = FALSE
)

# 兼容两种 characteristics 存储格式 (单列 \n 分隔 / 每字段一列)
get_char <- function(i, field) {
  cols <- grep("characteristics", colnames(meta), value = TRUE)
  # 格式 A: 列名就是 "cohort:chl", 值直接可用
  direct <- colnames(meta)[grepl(paste0("^", field, "\\s*:ch"), colnames(meta), ignore.case = TRUE)]
  if (length(direct) > 0) return(as.character(meta[i, direct[1]]))
  # 格式 B: 单列 characteristics_chl, 内含 "field: value" 多行
  for (v in unlist(lapply(meta[i, cols, drop = FALSE], as.character))) {
    if (is.na(v)) next
    for (line in strsplit(v, "\n", fixed = TRUE)[[1]]) {
      if (grepl(paste0("^", field, "\\s*:"), line, ignore.case = TRUE))
        return(sub(paste0("^", field, "\\s*:\\s*", ""), line, ignore.case = TRUE))
    }
  }
  NA_character_
}

design$cohort <- vapply(seq_len(nrow(meta))[m], get_char, character(1), field = "cohort")
design$age <- vapply(seq_len(nrow(meta))[m], get_char, character(1), field = "age")

```

```

design$sex    <- vapply(seq_len(nrow(meta))[m], get_char, character(1), field = "gender")

# ---- 6. cohort 数据质量处理 (记录进 cleaning log) ----
table(design$cohort, useNA = "ifany")
# 发现 2 个样本 cohort = "Diarrheaiarrhea", 疑似录入错误 (先核对 GSM 页确认属于 IBS-D)
# 决定: 核对后统一改写为 "Diarrhea" (写入 cleaning_summary.tsv)
design$cohort[design$cohort == "Diarrheaiarrhea"] <- "Diarrhea"
# 另有 1 个样本 cohort = NA: 从主对比中排除, 不自动猜测 (写入 cleaning log)

# ---- 7. 显式对齐 ----
rownames(design) <- design$sample_id
design <- design[colnames(counts), , drop = FALSE]
stopifnot(identical(colnames(counts), rownames(design)))

# ---- 8. 定义分析子集: T1 (timepoint=1) Diarrhea vs Healthy ----
analysis <- design$timepoint == "1" & design$cohort %in% c("Diarrhea", "Healthy")
cat("Analysis subset n =", sum(analysis), "\n")
counts_an <- counts[, analysis, drop = FALSE]
design_an <- design[analysis, , drop = FALSE]
table(design_an$cohort)          # Diarrhea 12 / Healthy 13

# ---- 9. 低表达过滤 (阈值=最小分组 n) ----
smallest_group <- min(table(design_an$cohort))    # 12
keep <- rowSums(counts_an >= 10) >= smallest_group
counts_f <- counts_an[keep, , drop = FALSE]
c(before = nrow(counts_an), after = nrow(counts_f)) # 64253 -> 约 24204
write.table(counts_f, "data_clean/GSE146853_counts_T1_filtered.tsv",
            sep = "\t", quote = FALSE, row.names = TRUE)

# ---- 10. QC 图: library size (按 cohort 着色) ----
lib <- colSums(counts_an)
o <- order(lib)
png("results/GSE146853_library_size.png", width = 1600, height = 1000, res = 180)
barplot(lib[o], las = 2, cex.names = 0.5, log = "y",
        col = ifelse(design_an$cohort[o] == "Diarrhea", "#EAA7B2", "#9BBBF4"),
        main = "GSE146853 library sizes (T1: Diarrhea vs Healthy, n = 25)",
        ylab = "Total raw counts (log10)")
legend("topleft", legend = c("Diarrhea (IBS-D)", "Healthy"),
      fill = c("#EAA7B2", "#9BBBF4"), cex = 0.8, bty = "n")
dev.off()

# 可选: PCA (需 DESeq2; 本周不强制)
# library(DESeq2)
# dds <- DESeqDataSetFromMatrix(counts_f, design_an, design = ~ cohort)
# vsd <- vst(dds, blind = FALSE)
# plotPCA(vsd, intgroup = "cohort")

# ---- 11. 清洗汇总 + 可复现信息 ----
qc <- data.frame(sample_id = names(lib), library_size = as.numeric(lib),
                cohort = design_an$cohort, patient_id = design_an$patient_id,
                timepoint = design_an$timepoint)
write.table(qc, "results/library_size_summary.tsv", sep = "\t",
            quote = FALSE, row.names = FALSE)
writeLines(capture.output(sessionInfo()), "sessionInfo.txt")

```


附录 1：week3_homework.R

R 脚本（修复版，可直接运行；完整说明见主文档附录 A）

```
# =====
# week3_homework.R — GSE146853（修正版，可直接运行）
# 修正说明（相对初版）：
#   1) read.delim 用 row.names = "GeneID"（不要用第 1 列 Chr），
#       并只保留 68 个 s_* 计数列（初版会把 6 列注释也当数据）。
#   2) 计数列名是 s_{患者}-{时间点}_GeneCount，不是 GSM 编号；
#       通过 meta$title 建立 列名 -> GSM 的显式映射（初版 intersect 会得 0）。
#   3) patient_id / timepoint 不在 characteristics 里，而在 title
#       （如 10007589-1）；characteristics 只有 age/bmi/gender/cohort。
#   4) 先把分析子集定义为 T1 的 Diarrhea vs Healthy，再算最小分组。
# =====
# 若 GEOquery 未安装才需执行（本机已装；勿运行 install，否则会触发 Update all/some/none
# if (!requireNamespace("BiocManager", quietly=TRUE)) install.packages("BiocManager")
# BiocManager::install("GEOquery", update = FALSE, ask = FALSE)
library(GEOquery)

dir.create("data_raw/GSE146853", recursive = TRUE, showWarnings = FALSE)
dir.create("metadata", recursive = TRUE, showWarnings = FALSE)
dir.create("results", recursive = TRUE, showWarnings = FALSE)
dir.create("data_clean", recursive = TRUE, showWarnings = FALSE)

# - - - - 1. GEO 研究对象（Series Matrix 主要取元数据） - - - -
gse_list <- getGEO("GSE146853", GSEMatrix = TRUE, getGPL = FALSE)
se <- gse_list[[1]]
if (inherits(se, "ExpressionSet")) {
  meta <- Biobase::pData(se)
} else {
  meta <- as.data.frame(colData(se))
}
cat("metadata rows:", nrow(meta), "\n")
print(colnames(meta)) # 先看有哪些列 (title / geo_accession / characteristics)

# - - - - 2. 补充文件清单 -> 下载计数矩阵 - - - -
# 注：getGEOSuppFiles() 的参数是 fetch 而非 fetch_files，仅下载即可
getGEOSuppFiles("GSE146853", baseDir = "data_raw")
count_file <- list.files("data_raw/GSE146853",
                        pattern = "GSE146853_GeneCount_raw\\.txt",
                        full.names = TRUE)

stopifnot(length(count_file) == 1)

# - - - - 3. 读入：GeneID 作行名；只留 68 个计数列 - - - -
counts_all <- read.delim(count_file, row.names = "GeneID",
                        check.names = FALSE, stringsAsFactors = FALSE)
counts <- counts_all[, grepl("^s_", colnames(counts_all))]
dim(counts) # 预期 64253 x 68

# - - - - 4. 计数性质核验（全部应通过） - - - -
stopifnot(all(as.matrix(counts) >= 0)) # 非负
stopifnot(all(as.matrix(counts) == round(as.matrix(counts)))) # 整数
stopifnot(sum(is.na(counts)) == 0) # 无缺失
stopifnot(anyDuplicated(rownames(counts)) == 0) # GeneID 唯一
range(counts) # 0 ~ 4184346

# - - - - 5. 从标题解析患者/时间点，从 characteristics 解析 cohort - - - -
cnt_title <- sub("_GeneCount$", "", sub("^s_", "", colnames(counts)))
m <- match(cnt_title, meta$title)
stopifnot(!any(is.na(m))) # 68/68 必须全部匹配

design <- data.frame(
  sample_id = colnames(counts),
  title = cnt_title,
  gsm = meta$geo_accession[m],
  patient_id = sub("-[12]$", "", cnt_title),
  timepoint = sub("[0-9]+-", "", cnt_title), # "1" 或 "2"
  stringsAsFactors = FALSE
)

# 兼容两种 characteristics 存储格式（单列 \n 分隔 / 每字段一列）
get_char <- function(i, field) {
```

```

cols <- grep("characteristics", colnames(meta), value = TRUE)
# 格式 A:列名就是 "cohort:ch1", 值直接可用
direct <- colnames(meta)[grepl(paste0("^", field, "\\s*:ch"), c
if (length(direct) > 0) return(as.character(meta[i, direct[1]]))
# 格式 B:单列 characteristics_ch1, 内含 "field: value" 多行
for (v in unlist(lapply(meta[i, cols, drop = FALSE], as.character
  if (is.na(v)) next
  for (line in strsplit(v, "\n", fixed = TRUE)[[1]]) {
    if (grepl(paste0("^", field, "\\s*:"), line, ignore.case =
      return(sub(paste0("^", field, "\\s*:\\s*"), "", line, ign
  }
}
NA_character_
}
design$cohort <- vapply(seq_len(nrow(meta))[m], get_char, character
design$age <- vapply(seq_len(nrow(meta))[m], get_char, character
design$sex <- vapply(seq_len(nrow(meta))[m], get_char, character

# - - - - 6. cohort 数据质量处理 (记录进 cleaning log) - - - -
table(design$cohort, useNA = "ifany")
# 发现 2 个样本 cohort = "Diarrheaiarrhea", 疑似录入错误 (先核对 GSM 页确认属于 IBS-D)
# 决定:核对后统一改写为 "Diarrhea" (写入 cleaning_summary.tsv)
design$cohort[design$cohort == "Diarrheaiarrhea"] <- "Diarrhea"
# 另有 1 个样本 cohort = NA:从主对比中排除, 不自动猜测 (写入 cleaning log)

# - - - - 7. 显式对齐 - - - -
rownames(design) <- design$sample_id
design <- design[colnames(counts), , drop = FALSE]
stopifnot(identical(colnames(counts), rownames(design)))

# - - - - 8. 定义分析子集: T1 (timepoint=1) Diarrhea vs Healthy - - - -
analysis <- design$timepoint == "1" & design$cohort %in% c("Diarrhea", "Healthy")
cat("Analysis subset n =", sum(analysis), "\n")
counts_an <- counts[, analysis, drop = FALSE]
design_an <- design[analysis, , drop = FALSE]
table(design_an$cohort) # Diarrhea 12 / Healthy 13

# - - - - 9. 低表达过滤 (阈值=最小分组 n) - - - -
smallest_group <- min(table(design_an$cohort)) # 12
keep <- rowSums(counts_an >= 10) >= smallest_group
counts_f <- counts_an[keep, , drop = FALSE]
c(before = nrow(counts_an), after = nrow(counts_f)) # 64253 -> 29753
write.table(counts_f, "data_clean/GSE146853_counts_T1_filtered.tsv",
  sep = "\t", quote = FALSE, row.names = TRUE)

# - - - - 10. QC 图: library size (按 cohort 着色) - - - -
lib <- colSums(counts_an)
o <- order(lib)
png("results/GSE146853_library_size.png", width = 1600, height = 1000)
barplot(lib[o], las = 2, cex.names = 0.5, log = "y",
  col = ifelse(design_an$cohort[o] == "Diarrhea", "#EAA7B2", "#9BBBF4"),
  main = "GSE146853 library sizes (T1: Diarrhea vs Healthy)",
  ylab = "Total raw counts (log10)")
legend("topleft", legend = c("Diarrhea (IBS-D)", "Healthy"),
  fill = c("#EAA7B2", "#9BBBF4"), cex = 0.8, bty = "n")
dev.off()

# 可选: PCA (需 DESeq2; 本周不强制)
# library(DESeq2)
# dds <- DESeqDataSetFromMatrix(counts_f, design_an, design = ~ cohort)
# vsd <- vst(dds, blind = FALSE)
# plotPCA(vsd, intgroup = "cohort")

# - - - - 11. 清洗汇总 + 可复现信息 - - - -
qc <- data.frame(sample_id = names(lib), library_size = as.numeric(lib[o]),
  cohort = design_an$cohort, patient_id = design_an$patient_id,
  timepoint = design_an$timepoint)
write.table(qc, "results/library_size_summary.tsv", sep = "\t",
  quote = FALSE, row.names = FALSE)
writeLines(capture.output(sessionInfo()), "sessionInfo.txt")

```

- # 说明：正式差分表达（D E S e q 2、配对 / 重复测量设计）留到 W e e k 4 ；
- # 本周作业到「过滤 + 一张 Q C 图 + 决策记录」为止。

附录 2 : c l e a n i n g _ s u m m a r y . t s v

清洗日志：记录 cohort 修正与排除决策

Item	Student decision / expected evidence
Dataset	GSE146853
Data type	Bulk RNA-seq gene-count matrix
Supplementary file	GSE146853_GeneCount_raw.txt.gz
Expected value type	Raw non-negative integer gene counts; verify
Matrix structure	Genes/features x samples; verify dimensions and
Missing values	Check <code>sum(is.na(counts))</code> ; should be reported, not
Duplicate genes	Check <code>anyDuplicated(rownames(counts))</code>
Duplicate samples	Check <code>anyDuplicated(colnames(counts))</code>
Metadata alignment	Explicitly match matrix columns to sample IDs
Primary comparison	IBS-D vs healthy controls at the first biopsy/
Covariates	Age, sex, BMI; patient ID/time point for repeated obse
Filtering	Keep genes with ≥ 10 counts in at least the size of the
Normalization	For later RNA-seq modeling, use count-aware methods
QC figure	Library-size plot generated by the R script
Important limitation	GSE146853 is longitudinal and contains repea

附录 3: ai_verification_log.txt

AI 使用与验证记录

Week 3 AI Verification Log

Prompt:

"Using the Week 3 courseware, help me reconstruct the GEO accessions."

AI advice accepted:

- Use GSE as the study-level identifier, GSM for sample-level metadata.
- Inspect the supplementary count file before analysis.
- Verify that RNA-seq count values are non-negative integers.
- Preserve subject ID and time point because the study is longitudinal.
- Define a specific contrast before any differential-expression analysis.
- Keep raw files unchanged and write cleaned data to separate paths.

Verification sources/tests:

1. NCBI GEO GSE146853 record: confirms Homo sapiens, bulk high-throughput RNA-seq.
2. Sample-level GEO record (e.g., GSM4407881): confirms tissue biopsies.
3. R checks in week3_homework.R:

```
all(counts >= 0)
all(counts == round(counts))
sum(is.na(counts))
anyDuplicated(rownames(counts))
anyDuplicated(colnames(counts))
identical(colnames(counts), metadata$sample_id)
```

Rejected advice:

- Do not assume that every sample is an independent biological replicate.
- Do not run DESeq2 on normalized TPM/FPKM values.
- Do not claim the QC results were executed until the R script has finished.

附录 4：library_size_summary.tsv

25 个 T1 样本文库大小明细 (QC 图数据)

sample_id	library_size	cohort	patient_id	timepoint
s_10007541-1_GeneCount	40113143	Healthy	10007541	1
s_10007544-1_GeneCount	52581059	Diarrhea	10007544	1
s_10007549-1_GeneCount	67614237	Healthy	10007549	1
s_10007554-1_GeneCount	69978701	Diarrhea	10007554	1
s_10007557-1_GeneCount	60366009	Healthy	10007557	1
s_10007559-1_GeneCount	57700574	Diarrhea	10007559	1
s_10007561-1_GeneCount	61599902	Diarrhea	10007561	1
s_10007562-1_GeneCount	62893029	Healthy	10007562	1
s_10007563-1_GeneCount	103439256	Healthy	10007563	1
s_10007564-1_GeneCount	44624454	Healthy	10007564	1
s_10007565-1_GeneCount	48750946	Diarrhea	10007565	1
s_10007566-1_GeneCount	68855041	Diarrhea	10007566	1
s_10007567-1_GeneCount	66486836	Diarrhea	10007567	1
s_10007569-1_GeneCount	52896134	Healthy	10007569	1
s_10007578-1_GeneCount	79532547	Diarrhea	10007578	1
s_10007581-1_GeneCount	65311465	Diarrhea	10007581	1
s_10007584-1_GeneCount	64128795	Diarrhea	10007584	1
s_10007587-1_GeneCount	84963740	Healthy	10007587	1
s_10007590-1_GeneCount	61891712	Diarrhea	10007590	1
s_10007594-1_GeneCount	75634277	Diarrhea	10007594	1
s_10007599-1_GeneCount	68578922	Healthy	10007599	1
s_10007603-1_GeneCount	48852943	Healthy	10007603	1
s_10007605-1_GeneCount	52537668	Healthy	10007605	1
s_10007613-1_GeneCount	81489965	Healthy	10007613	1
s_10007616-1_GeneCount	85167300	Healthy	10007616	1

附录 5 : sessionInfo.txt

R 会话信息 (可复现性记录)

R version 4.5.1 (2025-06-13 ucrt)

Platform: x86_64-w64-mingw32/x64

Running under: Windows 11 x64 (build 22631)

Matrix products: default

LAPACK version 3.12.1

locale:

[1] LC_COLLATE=Chinese (Simplified)_China.utf8

[2] LC_CTYPE=Chinese (Simplified)_China.utf8

[3] LC_MONETARY=Chinese (Simplified)_China.utf8

[4] LC_NUMERIC=C

[5] LC_TIME=Chinese (Simplified)_China.utf8

time zone: Etc/GMT-8

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] GEOquery_2.76.0 Biobase_2.68.0 BiocGenerics_0.54.0

[4] generics_0.1.4

loaded via a namespace (and not attached):

[1] rappdirs_0.3.3 tidy_1.3.2

[3] SparseArray_1.8.1 xml2_1.4.0

[5] lattice_0.22-7 hms_1.1.3

[7] magrittr_2.0.3 grid_4.5.1

[9] R.oo_1.27.1 jsonlite_2.0.0

[11] Matrix_1.7-3 R.utils_2.13.0

[13] GenomeInfoDb_1.44.2 limma_3.64.3

[15] httr_1.4.7 purrr_1.1.0

[17] UCSC.utils_1.4.0 XML_3.99-0.24

[19] httr2_1.2.1 abind_1.4-8

[21] cli_3.6.5 rlang_1.3.0

[23] crayon_1.5.3 XVector_0.48.0

[25] R.methodsS3_1.8.2 withr_3.0.2

[27] DelayedArray_0.34.1 S4Arrays_1.8.1

[29] tools_4.5.1 tzdb_0.5.0

[31] dplyr_1.2.1 GenomeInfoDbData_1.2.14

[33] SummarizedExperiment_1.38.1 curl_7.0.0

[35] vctrs_0.7.3 R6_2.6.1

[37] matrixStats_1.5.0 stats4_4.5.1

[39] lifecycle_1.0.5 S4Vectors_0.46.0

[41] IRanges_2.42.0 pkgconfig_2.0.3

[43] pillar_1.11.0 rentrez_1.2.4

[45] data.table_1.17.8 glue_1.8.0

[47] statmod_1.5.2 tibble_3.3.0

[49] GenomicRanges_1.60.0 tidyselect_1.2.1

[51] MatrixGenerics_1.20.0 readr_2.1.5

[53] compiler_4.5.1